

KOYTENDAG, Turkmenistan

WMO: 389150

Lat: 37.517N

Lon: 66.017E

Elev: 265

StdP: 98.18

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-3.6	-11.1	1.5	4.9	-8.5	1.9	6.5	12.6	14.8	11.6	14.6	1.6	270	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	41.8	20.0	40.4	19.8	39.2	19.6	23.0	35.4	21.7	35.8	20.8	35.7	4.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	29.7	17.0	12.5	27.5	15.6	11.4	26.7	69.5	35.6	64.5	35.7	61.3	35.8	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	9.8	8.2	-8.9	44.4	4.6	1.2	-12.2	45.3	-14.9	46.0	-17.5	46.6	-20.9	47.5		
			-9.1	24.3	4.5	2.3	-12.4	26.0	-15.0	27.4	-17.5	28.7	-20.8	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	6.0	8.1	14.1	20.3	26.2	30.8	32.2	30.0	24.8	18.5	11.5	6.8
	DBStd	10.23	5.36	5.64	5.33	4.87	3.65	2.68	2.31	2.62	3.15	4.29	5.00	4.69
	HDD10.0	418	142	93	22	2	0	0	0	0	0	2	39	119
	HDD18.3	1477	382	288	152	36	2	0	0	0	1	51	211	356
	CDD10.0	3765	19	38	149	310	502	624	687	621	443	265	85	22
	CDD18.3	1784	0	1	21	95	246	374	429	363	194	56	7	1
	CDH23.3	25408	0	4	188	964	3106	5782	6885	5380	2402	627	68	1
	CDH26.7	15311	0	0	45	381	1660	3721	4632	3417	1229	209	16	0
Wind	WSAvg	2.8	2.8	3.4	3.5	3.0	2.9	3.1	3.1	2.7	2.3	2.3	2.2	2.4
Precipitation	PrecAvg	194	33	32	41	30	15	1	0	0	1	3	20	27
	PrecMax	299	121	67	68	136	50	8	1	1	5	13	64	85
	PrecMin	103	4	8	4	0	0	0	0	0	0	0	2	0
	PrecStd	50	23	16	18	28	15	2	0	0	1	3	17	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	23.6	30.6	35.6	39.5	42.8	44.0	42.7	38.3	33.2	29.1	21.9
		MCWB	10.0	12.1	15.5	18.8	19.8	20.3	20.7	20.5	19.1	16.9	14.5	10.6
	2%	DB	18.2	20.8	27.8	32.8	37.5	41.0	42.3	40.6	36.5	31.0	25.1	19.3
		MCWB	8.9	10.7	14.0	17.7	18.9	19.9	20.1	20.4	18.8	15.9	13.3	9.4
	5%	DB	15.9	18.6	25.5	30.7	35.8	39.6	40.9	39.2	34.8	29.1	22.3	17.0
		MCWB	8.1	9.9	13.3	17.0	18.7	19.3	19.9	19.9	18.3	15.7	12.1	8.9
	10%	DB	13.8	16.6	23.1	28.7	34.1	38.3	39.6	37.7	32.9	26.9	20.0	14.5
		MCWB	7.5	9.1	12.4	16.3	18.1	19.1	19.6	19.4	17.6	14.9	11.5	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	13.4	16.7	21.7	24.0	23.2	23.9	25.4	23.8	20.3	15.6	12.1
		MCDB	17.1	19.9	25.0	30.3	33.4	38.8	37.4	36.8	31.6	26.5	25.2	17.5
	2%	WB	9.8	12.0	15.3	19.5	21.2	21.6	22.1	22.7	20.6	17.5	14.3	10.6
		MCDB	16.1	18.4	24.5	28.8	32.7	37.2	37.7	35.1	31.5	28.0	22.5	17.2
	5%	WB	8.9	10.9	14.3	18.3	19.9	20.7	21.2	21.2	19.1	16.2	13.1	9.4
		MCDB	14.7	16.6	22.8	27.8	32.0	36.4	37.4	35.9	32.5	27.1	20.3	15.5
	10%	WB	7.9	9.8	13.3	17.2	19.0	19.9	20.6	20.1	18.0	15.2	12.1	8.3
		MCDB	13.1	15.5	21.1	26.8	31.0	35.5	36.8	35.7	31.8	25.9	19.3	13.6
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.3	10.3	11.4	12.7	13.7	14.2	15.1	15.3	13.6	10.9	9.4
		MCDBR	12.0	11.8	12.9	13.8	14.7	15.4	15.7	16.4	17.0	15.6	14.4	13.1
		MCWBR	5.7	5.2	4.9	5.2	5.1	4.9	4.6	5.1	6.0	6.3	6.2	6.1
	5% WB	MCDBR	10.5	10.0	11.5	12.4	13.1	14.2	14.6	15.5	16.2	14.6	12.4	11.1
	MCWBR	5.6	4.9	5.1	5.5	5.7	5.4	5.4	5.9	7.0	6.7	6.2	5.9	
Clear-Sky Solar Irradiance	taub	0.393	0.441	0.500	0.513	0.509	0.507	0.492	0.465	0.443	0.477	0.452	0.399	
	taud	2.091	1.973	1.854	1.864	1.895	1.889	1.922	1.982	2.002	1.893	1.941	2.064	
	Ebn at Noon	753	763	756	776	788	788	796	808	800	711	671	709	
	Edh at Noon	120	153	190	198	196	197	189	174	161	163	139	116	
All-Sky Solar Radiation	RadAvg	2.18	3.01	4.29	5.73	7.17	8.11	7.92	7.22	6.04	4.29	2.71	2.06	
	RadStd	0.23	0.35	0.36	0.50	0.28	0.15	0.13	0.15	0.12	0.25	0.25	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	-0.83	-1.43	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	-0.30	-0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page